

Compton production and detection of dark photons

TEXONO: vacuum calculation, medium correction, and published limits

ROOT/C++

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Abstract

The tree-level production cross section passes independent matrix, Ward-identity, total-integral and massless-limit checks. Recalculating Park's Figure 1 does not reproduce the full published spectrum; the disagreement is displayed without fitting. For TEXONO we compare Park's quoted bound, the curve extracted from Danilov's PDF, an independent calculation with Park's conventions, and a homogeneous-medium calculation. The medium result includes an explicit signal-selection efficiency. No exclusion is claimed at material resonance or outside the stated small-mixing domain.

1 What is derived, and what is taken from sources

We reuse and recheck the project's v2 polynomial trace kernel. QED vertices, the electron propagator, spin completeness and two-body phase space are input identities. The compact trace, numerical convolution, inverse cross section and exclusion solver are implemented here. Gondolo–Raffelt (Park's Reference 15) supplies an independent total-cross-section check; it is not required to generate the differential kernel. Park supplies the empirical source and experimental inputs. Danilov supplies the medium probabilities and selection correction.

2 Production amplitude and trace

$$\gamma(k) + e(p) \rightarrow X(q) + e(p'), \quad p^2 = p'^2 = m^2, \quad k^2 = 0, \quad q^2 = M^2, \quad g = \text{diag}(1, -1, -1, -1). \quad (1)$$

$$\mathcal{L}_{\text{int}} = -e\bar{\psi}\gamma^\mu\psi(A_\mu + \epsilon X_\mu), \quad (2)$$

$$\mathcal{M} = \epsilon e^2 \bar{u}(p') \left[\gamma^\nu \frac{\not{p} + \not{k} + m}{s - m^2} \gamma^\mu + \gamma^\mu \frac{\not{p}' - \not{q} + m}{u - m^2} \gamma^\nu \right] u(p) \varepsilon_\mu \xi_\nu^*. \quad (3)$$

The two electron-exchange diagrams are both required. On-shell Dirac equations give zero upon contraction with either external vector momentum. Therefore the massive polarization sum can be replaced by the metric in the squared amplitude; the longitudinal term vanishes by current conservation. This does not discard a physical polarization.

$$\overline{|\mathcal{M}|^2} = \epsilon^2 e^4 F, \quad a = s - m^2, \quad b = u - m^2, \quad c = m^2, \quad d = M^2, \quad (4)$$

$$F = -2 \left(\frac{a}{b} + \frac{b}{a} \right) + 4(2c + d) \left[\frac{1}{a} + \frac{1}{b} + c \left(\frac{1}{a} + \frac{1}{b} \right)^2 - \frac{d}{ab} \right]. \quad (5)$$

The spin trace averages over two electron spins and two incoming photon polarizations, a factor of one quarter. It follows by expanding both diagrams and applying the even gamma-matrix trace identities. The code checks the polynomial identity, then evaluates the same amplitude using independent complex Dirac matrices and explicit physical polarizations.

3 Laboratory cross section and threshold

$$s = m^2 + 2mE_\gamma, \quad b = M^2 - 2mE_X, \quad E_{\gamma,\text{th}} = M + \frac{M^2}{2m}, \quad (6)$$

$$\lambda(s, m^2, M^2) = [s - (m + M)^2][s - (m - M)^2], \quad (7)$$

$$E_X^\pm = \frac{(E_\gamma + m)(s + M^2 - m^2) \pm E_\gamma \sqrt{\lambda}}{2s}, \quad (8)$$

$$\frac{d\sigma_{\gamma e \rightarrow Xe}}{dE_X} = \frac{\epsilon^2 e^4 (\hbar c)^2}{32\pi m E_\gamma^2} F, \quad E_X^- \leq E_X \leq E_X^+. \quad (9)$$

Outside the endpoints or below threshold the cross section is zero. Integrating the expression numerically agrees with the derived analytic primitive and Reference 15. At zero mass the integrated result is Klein–Nishina times the squared mixing. These are free-electron, tree-level statements; they do not establish that Compton scattering equals total material attenuation.

$$\Phi_X^{\text{Park}}(x) = \int_{\max(1 \text{ MeV}, E_{\text{th}})}^{40 \text{ MeV}} \frac{\Phi_\gamma(E)}{\sigma_{\text{KN}}(E)} \frac{d\sigma_{\text{prod}}(E, x)}{dx} dE, \quad \Phi_\gamma(E) = 0.58 \times 10^{18} \frac{P}{\text{MW}} e^{-E/(0.91 \text{ MeV})}. \quad (10)$$

The kernel enforces the additional angular support. Figure 1 uses 1 GW and the three original masses. Open points are vector coordinates extracted from the supplied Park PDF in v2, reused as source data. The original calculation is not reproduced in full; changing the source cutoff or fitting an overall factor is not a verified remedy. The gamma spectrum is an empirical fit above 0.2 MeV, while Park’s stated integration cutoff is 1 MeV.

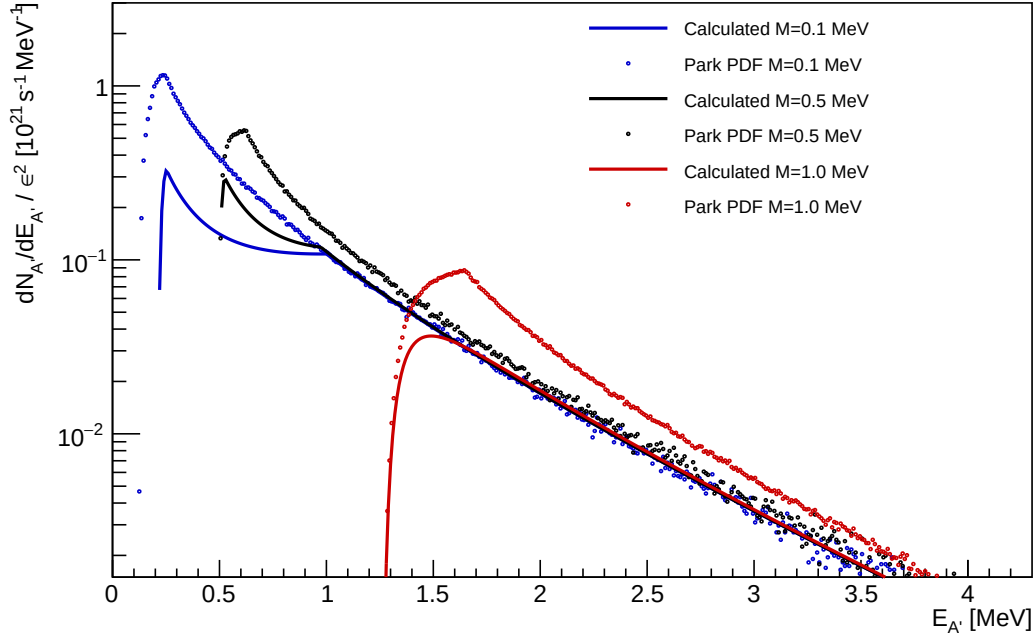


Figure 1: Recalculation versus Park Figure 1. Solid curves are computed; markers are extracted source data.

4 Detection by the inverse process

Time reversal gives the same invariant squared amplitude. For an unpolarized incident massive vector the average over three polarizations adds the factor two thirds relative to production. For incident energy E , momentum p_X , and outgoing photon energy w ,

$$p_X = \sqrt{E^2 - M^2}, \quad s = m^2 + M^2 + 2mE, \quad a = M^2 + 2mE, \quad b = -2mw, \quad (11)$$

$$w_{\pm} = \frac{(s - m^2)(E + m \pm p_X)}{2s}, \quad (12)$$

$$\frac{d\sigma_{Xe \rightarrow \gamma e}^{\text{unpol}}}{dw} = \frac{2}{3} \frac{\epsilon^2 e^4 (\hbar c)^2 F}{32\pi m p_X^2}. \quad (13)$$

The inverse massless limit is two thirds of Klein–Nishina. This is appropriate for the Park convention. It is not the factor used for oscillation-produced transverse states. Full containment gives $E_{\text{vis}} = E$ after combining electron and photon energy. A real detector needs a response function; replacing it by full containment is an explicit approximation.

5 Medium production and conversion

$$m_\gamma^2 = 4\pi\alpha n_e/m, \quad (14)$$

$$(\Delta m^2)^2 = (M^2 - m_\gamma^2)^2 + 2\epsilon^2 M^2 (M^2 + m_\gamma^2), \quad (15)$$

$$P_{\gamma \rightarrow X} = \frac{\epsilon^2 M^4}{(\Delta m_R^2)^2 + E^2 \Gamma_R^2}, \quad \Gamma_R = \frac{\hbar c}{L_{\text{abs}}}, \quad (16)$$

$$P_{X \rightarrow \gamma} = \frac{\epsilon^2 M^4}{(\Delta m_D^2)^2}, \quad \Phi_X^{\text{med}}(E) = \Phi_\gamma(E) P_{\gamma \rightarrow X}(E). \quad (17)$$

These are Danilov Eqs. (5), (6), and (9), not all-orders exact probabilities. The earlier implementation incorrectly used 4 instead of 2 in the splitting and treated the detector expression as valid at resonance. Equation (9) explicitly neglects absorption away from resonance. We omit a conservative ten-percent mass window around either plasma mass. A finite-size, absorbing, inhomogeneous detector calculation is needed within it. We set both plasma masses to 20 eV and the reactor absorption length to 10 cm, following the paper’s illustrative choices. At masses well above the plasma scale both probabilities approach the squared mixing. At much smaller masses their product scales as $\epsilon^4 (M/m_\gamma)^8$ when damping is negligible. The direct oscillation source preserves photon energy; the Compton-convolved Park source redistributes it. They are alternative source prescriptions and are not multiplied together.

6 Event rate, statistics, and comparisons

$$N_e = M_{\text{det}}[\text{kg}] 10^3 N_A Z/A, \quad (18)$$

$$N_{\text{med}} = \frac{N_e T \eta}{4\pi R^2} \int_{3 \text{ MeV}}^{8 \text{ MeV}} \Phi_\gamma(E) P_{\gamma \rightarrow X}(E) P_{X \rightarrow \gamma}(E) \sigma_{\text{KN}}(E) dE, \quad (19)$$

$$N_{\text{med}}(\epsilon_{95}, M) = 195.7, \quad 0 < \epsilon_{95} \leq 0.01. \quad (20)$$

Inputs are 2.9 GW, 28 m, 187 kg of CsI, 160 live days, $Z = 108$ and $A = 259.8099219$ g/mol. The measured excess quoted by Danilov is 30.7 ± 100.6 events. Their one-sided cap is 195.7 selected

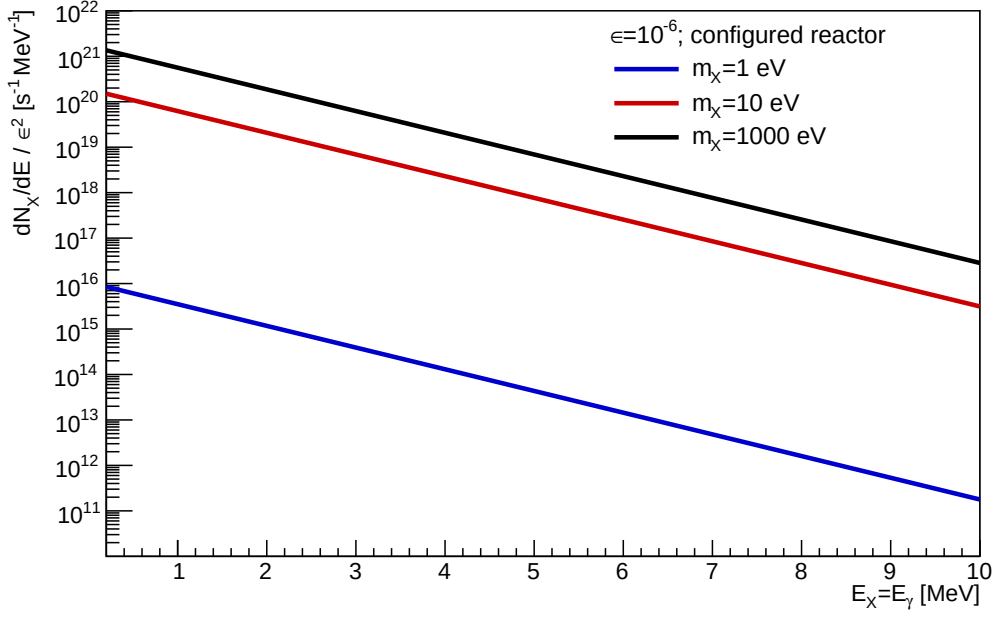


Figure 2: Medium source at the configured reactor power. The normalization is evaluated at $\epsilon = 10^{-6}$ and divided by its square.

events. Their assumed anti-Compton suppression corresponds to $\eta = 1/6$, approximately 1174 preselection events. It is an estimate, not a measured dark-photon efficiency. We also show unit efficiency to expose its effect. The earlier report used the selected-event cap with unit efficiency only and therefore omitted this correction. The Park-convention recalculation uses the convolved source, massive inverse cross section, unit efficiency and 1.96(100.6) events. Park labels this 95% CL; for a centered Gaussian it corresponds to a 97.5% one-sided cap. It must not be silently equated with the Danilov one-sided convention. The red line is Park's quoted 2.1×10^{-5} bound, not a newly digitized data set. The black line is extracted directly from Danilov's vector PDF. The independent medium calculation is not tuned to either line. Differences remain because the published normalization, source prescription and detector acceptance are not fully reproduced.

7 Validity and actual verification

The production kernel has exact two-body massive kinematics within tree-level free-electron QED. The medium detector model uses relativistic transverse modes, $M \ll m_e, E$, and is reported only up to 10 keV for a 3–8 MeV ROI. Plasma masses and absorption are constant inputs. Material attenuation, atomic binding, rescattering, escape of photons, finite geometry, and energy-dependent selection are not simulated. Thus the medium limit is conditional on this response and source model; it is not a validated detector-level TEXONO exclusion. The executed checks include independent matrix traces and Ward identities, the Reference 15 total, analytic integration, both massless scattering limits, the corrected splitting coefficient, root closure, doubled quadrature resolution, and rejection of invalid domains. Every plotted curve is checked for finite in-frame points and its presence in the canvas at export. The exported images are inspected visually. The original full Park spectrum and Danilov numerical curve are not reproduced by the independent prediction; their differences are visible.

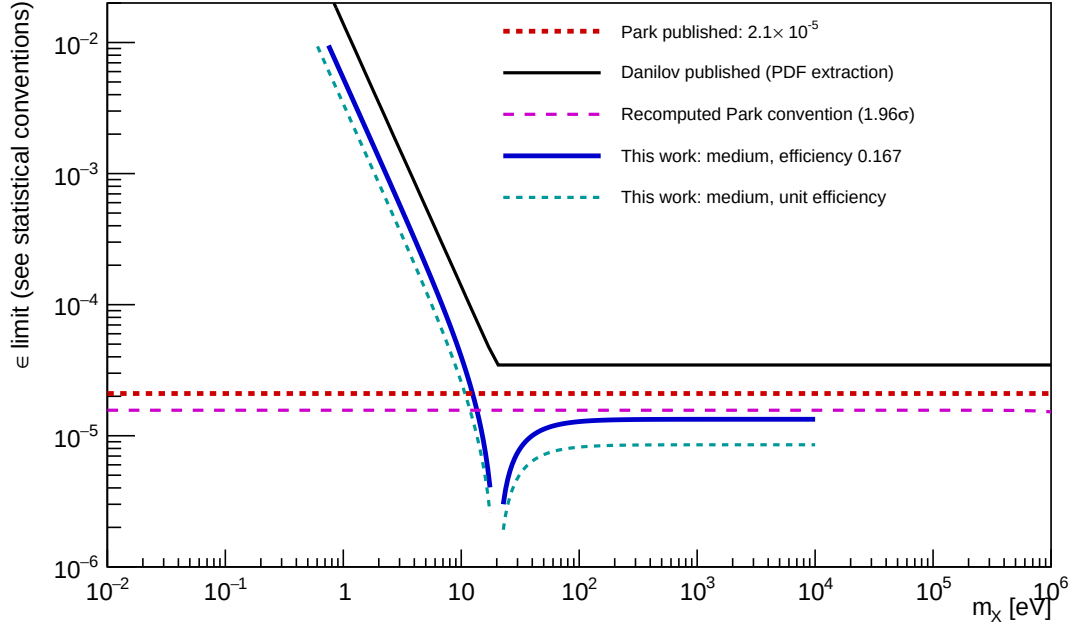


Figure 3: TEXONO-only comparison. Gaps near 20 eV mark the unclaimed resonance region. Medium curves end at 10 keV; the exact massive Park-convention calculation extends to 1 MeV.

8 Decay survival and the requested sub-MeV sensitivity

This execution adds propagation to the checked kernels. It does not derive the electron loop. Izaguirre, Krnjaic and Pospelov, Phys. Rev. D 92, 095014 (2015), Eqs. (9)–(11), distinguish absorption after survival from decays inside a detector. The source is available locally as `papers/park_reproduction/reference20.pdf`. Their electron-pair decay channel is closed for the masses considered here. We assume no lighter hidden particles and use the three-photon approximation quoted in Park Eq. (4):

$$\Gamma_{3\gamma}^{\text{approx}} = 2.16 \times 10^{-16} e^4 \epsilon^2 \frac{M^9}{m_e^8}, \quad (21)$$

$$\ell(E, M, \epsilon) = \frac{\hbar c \sqrt{E^2 - M^2}}{M \Gamma_{3\gamma}}, \quad S(E) = \exp[-R/\ell(E)], \quad (22)$$

$$N_{\text{vac}} = \frac{N_e T \eta}{4\pi R^2} \int_{E_{\text{min}}}^{E_{\text{max}}} \Phi_X^{\text{Park}}(E) S(E) \sigma_{X e \rightarrow \gamma e}(E) dE. \quad (23)$$

Both source and scattering include their squared mixing. For the medium model, insert the same survival factor in the earlier medium event integral; do not multiply the two alternative production models. The exponent uses the exact momentum instead of setting the speed to unity. For a detector chord L , a decay signal would contain $S(1 - e^{-L/\ell})$ and a separate acceptance; we do not count those events as Compton absorption. A detector geometry and decay selection are not provided.

The three-photon width is a low-mass expansion, controlled for $M \ll m_e$. Extending it to the upper endpoint below 1 MeV is a diagnostic, not a checked exact-loop prediction. The table records the largest propagation exponent at the vacuum boundary. This small approximate exponent is not an uncertainty bound on the exact width. Invisible decays, if allowed by extra

dark particles, would require a different model.

Largest approximate optical depth	1.0975×10^{-11}
Upper scan mass (MeV)	0.999

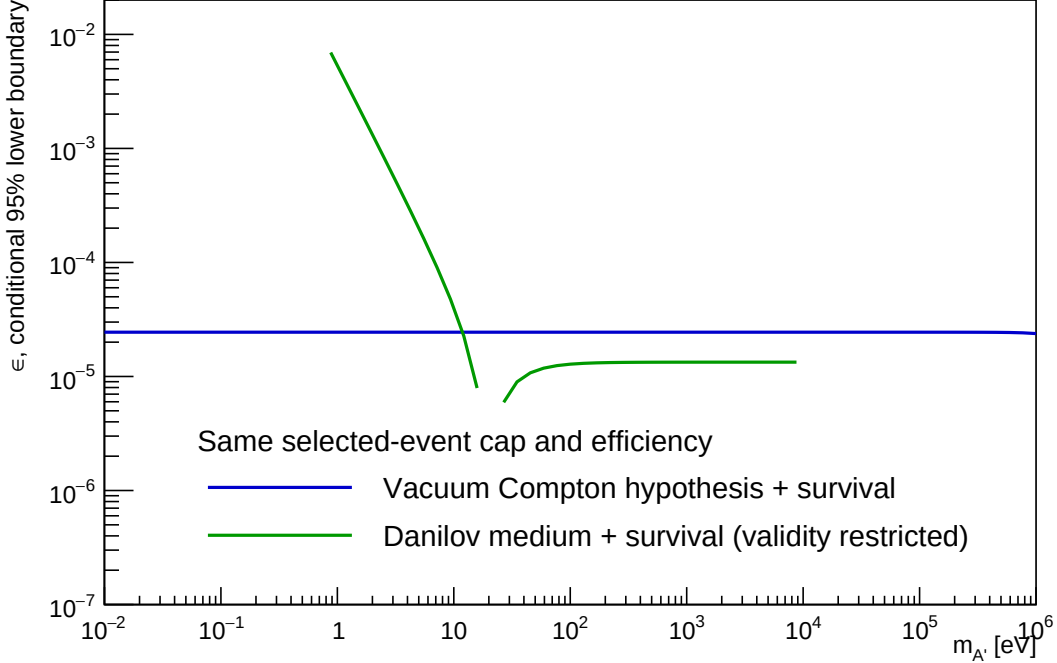


Figure 4: Same selected-event cap and efficiency in both hypotheses. The vacuum curve illustrates the massive Compton model; it is not a valid low-mass material exclusion. The medium curve is restricted to its relativistic validity range and excludes the resonance window.

9 Reproduction status and reuse

The only PDF in `papers_init` is Park. Its Figure 1 is recalculated and compared with its vector paths: the heavier-mass spectra disagree, so the full figure is not reproduced. Figure 2 combines TEXONO, NEOS and external astronomical, cosmological and laboratory exclusions. We calculate the TEXONO component, which also disagrees with the printed bound. The original external data and the private NEOS background analysis are not supplied; the complete Figure 2 is not reproduced. Showing the published page or digitizing it would not change this status.

Run `bash work/reactor_sensitivity/run.sh` from the project root. Edit `config.conf` for power, baseline, detector mass, exposure, composition, ROI, efficiency, medium parameters and the event cap. Supply a separately justified `N95_events` for a different likelihood, or use the documented Gaussian excess/error option. Background is represented by the excess and its uncertainty, not by a simulated spectrum. The current sub-MeV scan requires an ROI lower edge above 1 MeV. General lower-energy detectors need atomic and response modeling before reuse.

New C++ functions implement decay propagation, cached event integration and the first upward crossing of the event cap. ROOT supplies quadrature and rendering; existing project code supplies the checked trace and source convolution. No new microscopic amplitude is claimed. The

executed checks cover scaling, zero-distance and zero-mixing survival, analytic root recovery, absent roots, quadrature convergence and event-cap closure. Unchecked: the exact near-threshold loop, reactor transport, detector response, polarization transport in a Compton-produced beam, raw TEXONO likelihood and the full published exclusion compilation.

References

- [1] H. K. Park, Phys. Rev. Lett. 119, 081801 (2017), local source papers/darkphoton_v3/park2017.pdf, Eqs. (1)–(7), Figs. 1–2.
- [2] M. Danilov, S. Demidov and D. Gorbunov, arXiv:1804.10777 (2019), local source papers/darkphoton_v3/danilov2019.pdf, Eqs. (5), (6), (9), Fig. 1 and TEXONO discussion.
- [3] P. Gondolo and G. Raffelt, Phys. Rev. D 79, 107301 (2009), vector appendix, local source papers/darkphoton_v3/reference15.pdf.